

# **SIMATS ENGINEERING**

## **Assessment Tool 2**

**Course: Computer Vision with OpenCV (DSA0202)**

**Scenario-Based**

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### **Question 1: Crack Detection on Concrete Surfaces Using Drone Images**

#### **A. Image Preprocessing Pipeline**

##### **Step 1: Illumination Correction**

**Method:** Adaptive Histogram Equalization (CLAHE) or Homomorphic Filtering

**Purpose:**

- Removes uneven lighting caused by sunlight and shadows.
- Makes crack visibility more consistent.

**Justification:**

Drone images captured outdoors often have varying illumination. Correcting brightness ensures cracks remain visible in all regions.

##### **Step 2: Noise Reduction**

**Method:** Gaussian Blur or Median Filter

**Purpose:**

- Removes Gaussian noise caused by camera vibration and environmental conditions.
- Preserves important crack structures.

**Justification:**

Noise can create false edges during crack detection. Noise reduction improves detection accuracy.

##### **Step 3: Contrast Enhancement**

**Method:** CLAHE (Contrast Limited Adaptive Histogram Equalization)

**Purpose:**

- Enhances fine cracks.
- Improves local image contrast.

**Justification:**

Concrete cracks are usually thin and dark. Increasing contrast makes them easier to detect.

##### **Step 4: Image Normalization**

**Method:** Min-Max Normalization

**Purpose:**

- Converts pixel values into a common intensity range (0–255 or 0–1).

**Justification:**

Normalization ensures consistent processing regardless of image brightness.

**Complete Preprocessing Pipeline**

Drone Image



Illumination Correction



Gaussian Noise Removal



Contrast Enhancement (CLAHE)



Image Normalization



Ready for Crack Detection

**B. Suitable Edge Detection Technique****Selected Method: Canny Edge Detector****Why Canny?**

- Detects very thin cracks accurately.
- Removes noise before edge detection.
- Uses double thresholding.
- Produces continuous edges.

**Comparison with Sobel Edge Detector**

| Canny                     | Sobel                   |
|---------------------------|-------------------------|
| Better noise suppression  | Sensitive to noise      |
| Detects thin cracks       | Misses very fine cracks |
| Double threshold          | Single gradient         |
| Produces continuous edges | Broken edges            |

**Conclusion**

Canny Edge Detection is more suitable because it detects fine cracks with high accuracy while minimizing false edges.

### **C. Post-Processing Technique**

#### **Method: Morphological Closing**

##### **Operations:**

1. Dilation
2. Erosion

##### **Purpose**

- Connects broken crack segments.
- Fills small gaps.
- Produces continuous crack lines.

##### **Result**

Broken Crack



Morphological Closing

## **Question 2: Human Motion Tracking Using Security Camera**

### **A. Lucas-Kanade Optical Flow**

Lucas-Kanade Optical Flow estimates the movement of feature points between consecutive video frames.

#### **Working Steps**

1. Detect feature points.
2. Compare neighboring pixels in consecutive frames.
3. Calculate displacement vectors.
4. Estimate movement direction and speed.

#### **Advantages**

- Fast computation.
- Suitable for real-time tracking.
- Works well for small motions.

### **B. Aperture Effect**

#### **Definition**

The aperture effect occurs when only a small portion of an object's edge is visible, making the true motion direction ambiguous.

## **Effects on Uniform Regions**

Examples:

- Plain walls
- Smooth floors
- Clothing

These regions have very few texture features.

As a result:

- Motion cannot be estimated accurately.
- Optical flow vectors become incorrect or unstable.

## **C. Sudden Change in Speed or Direction**

When a person suddenly changes movement:

- Optical flow vectors change direction rapidly.
- Vector magnitude increases or decreases.
- Tracking may lose the target.

## **Improvement Method**

### **Pyramidal Lucas-Kanade Optical Flow**

Advantages:

- Detects large movements.
- Handles sudden motion changes.
- Improves tracking accuracy.

## **Question 3: Vehicle Tracking at Traffic Junction**

### **A. Optical Flow for Vehicle Tracking**

Optical flow estimates how pixels move between consecutive video frames.

#### **Process**

1. Detect vehicle features.
2. Compare consecutive frames.
3. Calculate motion vectors.
4. Estimate:
  - Direction
  - Speed

- Vehicle movement

### **Example**

Frame 1

Car →

Frame 2

Car →

Motion Vector →

Speed = Distance / Time

### **B. Two Challenges**

#### **Challenge 1: Occlusion**

When two vehicles overlap,

Effects:

- Vehicle boundaries disappear.
- Motion vectors become mixed.
- Tracking accuracy decreases.

#### **Challenge 2: Shadows**

Vehicle shadows move differently from the vehicles.

Effects:

- False optical flow vectors.
- Incorrect vehicle speed.
- Wrong movement direction.

### **C. Improvement Method**

#### **Method: Kalman Filter with Optical Flow**

**Why?**

The Kalman Filter predicts the next vehicle position even when:

- Vehicles overlap.
- Vehicles stop suddenly.
- Vehicles change direction.

#### **Advantages**

- Smooth tracking.

- Handles temporary occlusion.
- Predicts missing positions.
- Improves reliability.

### Final Summary

| Question                      | Best Technique                                                    |
|-------------------------------|-------------------------------------------------------------------|
| Crack Detection Preprocessing | Illumination Correction → Gaussian Filter → CLAHE → Normalization |
| Crack Edge Detection          | Canny Edge Detector                                               |
| Crack Post-processing         | Morphological Closing                                             |
| Human Motion Tracking         | Lucas-Kanade Optical Flow                                         |
| Large Motion Handling         | Pyramidal Lucas-Kanade                                            |
| Vehicle Motion Estimation     | Optical Flow                                                      |
| Vehicle Tracking Improvement  | Kalman Filter + Optical Flow                                      |